

Work and Practical Session: Machine Learning

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Session 04:

Supervised Learning – CNN

Exercise 1 – Manual 2D Convolution Calculations, Feature Maps and Hyperparameter Analysis

Consider the following 7×7 grayscale image:

```
0 0 0 0 0 0 0
0 1 1 1 0 0 0
0 1 4 1 0 0 0
0 1 1 1 0 0 0
0 0 0 0 2 2 0
0 0 0 2 5 2 0
0 0 0 0 0 0 0
```

Apply the following operations successively:

1. **Vertical edge detection filter** (3×3 kernel, no padding, stride=1):

```
1 0 -1
1 0 -1
1 0 -1
```

a) What is the expected output size? Justify using the standard convolution output size formula.

b) Compute the resulting **feature map** values.

2. Now apply a horizontal edge detection (padding= « same », stride=1) on the same input image:

```
1 1 1
0 0 0
-1 -1 -1
```

a) How much zero-padding (number of zeros added on each side) is required?

b) Compute the central 4 values of the resulting feature map (rows 3–4, columns 3–4).

3. Apply a **2×2 max-pooling** layer with stride 2 and no padding to the feature map obtained in question 1.
 - What is the output size?
 - Briefly explain why max-pooling is generally preferred over average-pooling in modern CNNs for image recognition tasks.
4. We want the output of a Conv2D layer to have spatial size exactly 32×32.

Input: 36×36×3 - Kernel size: 5×5 - Stride: 1

Which padding should be used?

Exercise 2 – Designing and Analyzing Small CNN Architectures

We want to build a very lightweight CNN to classify 32×32×3 images (CIFAR-10 or simplified SVHN style) with as few parameters as possible, while aiming to exceed 60–65% accuracy.

Part A – Parameter Counting

Complete the following table for these three architectures (all use ReLU after each convolution, followed by Global Average Pooling + a final Dense-10-softmax layer):

Architecture	Layers	Total number of parameters (approx.)	Feature map size before GAP
A	Conv-32 (3×3) → Conv-64 (3×3) → MaxPool 2×2 → Conv-128 (3×3) → GAP	?	?
B	Conv-16 (5×5) → MaxPool 2×2 → Conv-32 (5×5) → MaxPool 2×2 → GAP	?	?
C	Conv-8 (3×3, stride=2) → Conv-16 (3×3) → Conv-32 (3×3) → GAP	?	?

Hints:

- Use **same** padding for all convolutions
- For simplicity, you may assume no bias terms

Part B – Discussion Questions

1. Among A, B and C, which architecture is most likely to **underfit**? Why?

2. Which one is most likely to **overfit** on a small dataset (~10,000 images)?
3. If you are constrained to stay under **15,000 parameters**, which architecture(s) would you keep? Suggest one realistic modification to improve accuracy without significantly increasing the parameter count.
4. Explain why using **Global Average Pooling** instead of **Flatten + Dense(512)** is especially advantageous in very lightweight networks.

Part C – Open question (bonus)

Propose your own mini-CNN (maximum 4–6 layers) for CIFAR-10 with **fewer than 30,000 parameters** that should theoretically be capable of exceeding 70% accuracy (without data augmentation or advanced regularization). Justify your design choices (filter sizes, strides, number of filters, pooling strategy, etc.).